

- Q.30 Explain the working of step-up chopper with the help of circuit diagram.
- Q.31 Explain the working principle of dual converter.
- Q.32 How the speed of a DC motor can be controlled using chopper?
- Q.33 Write a short note on Maintenance of storage devices.
- Q.34 Write any five differences between on line UPS and off line UPS.
- Q.35 Explain illumination control circuit using TRIAC.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the different methods of triggering of an SCR. Explain in detail.
- Q.37 What are the storage devices? Explain in detail about the maintenance of batteries.
- Q.38 Explain cyclo-converter controlled AC drives with the help of suitable diagram. Also write advantages and disadvantages of cyclo-converter.

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5th Sem / Elect, Power Station Engg., Elect & Eltx Engg. Subject:- Industrial Electronics and Control of Drives

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In normal operation of an SCR, gate is _____ w.r.t cathode.
- a) Positive b) Negative
- c) At zero potential d) None of the above
- Q.2 How many terminals are there in a Uni-junction transistor?
- a) 1 b) 2
- c) 3 d) 4
- Q.3 Choppers converter
- a) AC to DC b) DC to AC
- c) DC to DC d) AC to AC
- Q.4 How many diodes are depicted in the DIACs symbol?
- a) Four b) three
- c) Two d) One
- Q.5 Which terminal does not belong to the SCR
- a) Anode b) Base
- c) Gate d) Cathode

- Q.6 SCRs are connected in parallel to fulfill the _____ demand.
- a) High voltage b) High current
c) size d) efficiency
- Q.7 Static UPS requires
- a) Only Rectifier
b) Only Inverter
c) Both Rectifier & Inverter
d) None of the above
- Q.8 The thyristor turn-off requires that the anode current
- a) falls below the holding current
b) falls below the latching current
c) falls above the holding current
d) falls above the latching current
- Q.9 In the SCR structure the gate terminal is located
- a) near the anode terminal
b) near the cathode terminal
c) in between the anode & cathode terminal
d) None of the above
- Q.10 The single-phase bridge type cycloconverter uses _____ number of SCRs.
- a) Four b) Eight
c) Six d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Duty cycle of chopper is given by _____
- Q.12 Draw symbol of DIAC.

- Q.13 Which material is used for making heat sinks.
- Q.14 Define commutation.
- Q.15 Define Latching current.
- Q.16 Write full form of CVT.
- Q.17 Which method is used to control the speed of DC shunt motor above normal speed
- Q.18 Define Cycloconverter.
- Q.19 List any one application of DIAC.
- Q.20 Expand UJT.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Draw the VI characteristics of DIAC and explain its working.
- Q.22 Explain working of UJT as Relaxation oscillator.
- Q.23 List five advantages of AC drive over DC drive.
- Q.24 Explain the working of single-phase half controlled full wave rectifier with resistive load.
- Q.25 Explain the working principle of series inverter with the help of circuit diagram.
- Q.26 Draw the symbol and constructional layer diagram of TRIAC. Draw its VI characteristics. Also write the different modes of conduction of a TRIAC.
- Q.27 Explain the working of an SCR using its layer diagram in brief.
- Q.28 Explain the series operation of thyristor.
- Q.29 Differentiate between Latching current and holding current.